

PADMANAVA KARMAKAR

Flutter Developer

My Contact

karpadmanava16@gmail.com

+91-7908767307

linkedin.com/in/padmanava-karmakar-5069641b6

Ahmedabad

Technical Skills

- Flutter & Dart
- BloC, Cubit, GetX, Provider, Riverpod
- REST APIs
- Clean architecture, MVVM
- Firebase, Supabase, SQFlite, Isar, etc.
- Github
- Deployement
- Animations
- App performance optimisation
- Google maps, OLA maps, Mapbox.
- AI tools like Cursor, Claude code, etc.

Soft Skill

- Effective communication
- Team player
- Empathy
- Problem-solving
- Adaptibility

Education Background

- Parul University
  Master of Computer Application(CGPA : 8.3)
  Completed in 2022
- Krishnath College
  B.Sc Physics(Hons.) (55.6%)
  Completed in 2019
- Kandi Raj High School(H.S)
  12 th , WBCHSE (88%)
- Kandi Raj High School(H.S)
  10 th , WBBSE (90.28%)

Languages

- Hindi
- English
- Bengali
- Gujarati

About Me

Strategic Flutter Developer with 4.5 years of experience shipping high-performance, cross-platform applications for iOS and Android. Expert in bridging the gap between Clean Architecture and rapid prototyping through AI-augmented development. Proven track record of accelerating the SDLC by 10x—transforming complex product requirements into polished, production-ready UIs and scalable codebases. Passionate about "Vibe Coding": where creative flow meets rigorous engineering to deliver apps that feel truly native.

Professional Experience

Bacancy Technology, Ahmedabad

Software Engineer
Sept 2023 - Present

Radixweb Pvt Ltd, Ahmedabad

Associate Software Engineer
Sept 2022 - August 2023

Radixweb Pvt Ltd

Trainee Software Engineer
Feb 2022 - Aug 2022

Projects

The Word

A comprehensive social impact mobile application that connects donors, charities, and communities through secure real-time communication, donation management, and community engagement features. The app facilitates charitable giving while building meaningful connections between users and organizations.

Team: 3 developers(now 2)
Position: Software Engineer

Technical Stack

- **Frontend**: Flutter/Dart with Material Design 3
- **Backend**: Firebase (Firestore, Auth, Storage, Cloud Functions)
- **Security**: RSA/AES encryption, Flutter Secure Storage
- **Payments**: Stripe integration
- **Search**: Typesense search engine
- **AI**: Google Gemini AI integration
- **Notifications**: Firebase Cloud Messaging
- **State Management**: Provider pattern with Riverpod
- **Media**: Image/video compression, caching, and processing

Key Features & Technical Achievements
End-to-End Encryption (E2EE) Chat System

- Implemented secure real-time messaging with RSA/AES encryption for text, images, and video files
- Built custom E2EE service using PointyCastle and Encrypt libraries for military-grade security
- Integrated secure key management with Flutter Secure Storage and Firebase Firestore
- Enabled optional encryption toggle for user privacy preference.

Donation & Payment Processing
Integrated Stripe payment gateway for secure credit card transactions
Built comprehensive donation tracking and management system
Implemented corporate matching functionality for enhanced giving impact
Created donation history, receipts, and impact tracking features.

Community & Charity Management
Developed full CRUD operations for charity and community creation/management
Built search functionality with Typesense integration for fast, relevant results
Implemented photo/video sharing, event creation, and member management
Created moderation system with admin controls and content filtering.

Real-time Communication
Built real-time chat system with Firebase Firestore and Cloud Functions
Implemented push notifications using Firebase Cloud Messaging
Created conversation management with unread message tracking
Added media sharing (images, videos) with compression and encryption.

Advanced Search & Discovery
Integrated Google Places API for location-based charity discovery
Built semantic search with Typesense for intelligent content discovery
Implemented filtering by causes, location, and charity type
Created personalized recommendations based on user preferences.

Gamification & Engagement
Built achievement/badge system to encourage user engagement
Implemented leaderboards and impact tracking
Created adventure/challenge features for community participation
Added reward system for charitable activities

● Rider Buddy

Designed and developed a real-time group cycling application featuring precise location sharing with up to 10m accuracy, QR-based group management, and live member status tracking. Implemented intelligent route planning and dynamic meeting point suggestions for late joiners, along with a live map experience using custom markers, polylines, and compass navigation. Built mid-journey detection, push notifications, and offline data synchronization, while optimizing Firebase listeners to support up to 1,000 concurrent users. Achieved efficient battery usage, and high-performance map rendering through clustering and caching techniques.

● DigInn

It is a restaurant application based on healthy foods. Users can order from all the Dig restaurants available according to their convenience, they can choose either pickup or delivery. Users can get every detail of the food they are going to order like the calories, protein, and other nutritional info, they can customize their meals from a large variety. This app works on the reward system. Users are divided into three tiers bronze, silver, and gold tier and the reward system for the tiers are different, also sign up reward, birthday rewards etc. are there as well.

Team: 2 developers, a QA, and a Tech lead
Position: Software developer

Tools: Getx(state management), dio(API calling), retrofit, thanx SDK for deep links, firebase(remote config, analytics, and notifications), isar(database), Clean Architecture.

Responsibilities:
- Implemented the application UI and developed reusable custom widgets to speed up application development
- Integrated third-party libraries to interact with the back-end API
- Designed dynamic and complex functionality and developed the application using the GetX state management focusing on reactive state management including navigations.
- Provided light theme and dark theme, common text style classes.
- Handling of widgets state properly to work with background data updates
- unit test cases

● Goldbank Driver

This is a delivery app used only for registered drivers for GoldBank located in London. The driver can log in and check their assigned collections, start a collection, collect the parcel from the customer, and drop the collection in the warehouse. The whole journey is tracked with Mapbox API and reflected in the admin panel as well. The app was previously built but had to be revamped with a whole new design and some changes in flow.

Tools: Riverpod(state management), Mapbox API
Team: 2 developers, a QA, and a Manager
Position: Software Engineer

● Goldbank Live & Sell Gold

Goldbank Live is an app like a live jewelry showroom where you can check all the new products of Goldbank. Users can inquire about products, also can check the different cryptocurrency and Foreign exchange rates as well. In this project, I worked a few months on maintenance like some UI changes or implementing API or bug fixing.

Sell Gold app is for customers who are willing to sell jewelry to Goldbank. Users can select Silver, Platinum, or Gold products can lock in the current price, and can avail the service. Here they can select premium service where Goldbank Driver comes to work. I worked on the maintenance of this project too.

Tools: GetX(state management)
Position: Software Engineer

● News App

This is a simple news application where the latest news headlines are shown on the top, below there different categories are present like business, entertainment, science, etc. You can also filter by country news from the country picker. There is a search page for searching with any keyword, suggestion box is available like Google, geopolitics etc hot topics.

Tools: Cubit(state management), dio(API calling), NewsAPi.org APIs, Clean Architecture.

AstraQuant

AstraQuant is an AI-driven trading intelligence app designed for the Indian stock market (NSE/BSE). It delivers ML-powered stock predictions, real-time sentiment analysis from news sources, portfolio tracking, and market regime detection—helping retail investors make smarter, data-backed decisions. Features include drift alerts for model monitoring, SHAP-based explainability so users understand why a prediction was made, and coverage across stocks, ETFs, mutual funds, and commodities.

Tools: Flutter (Riverpod, GoRouter, fl_chart) on the frontend and FastAPI + Python on the backend. The ML engine combines XGBoost, LightGBM, and custom expert models (technical, fundamental, sentiment, regime) into a meta-learner ensemble, with SHAP for transparency. Real-time data streams via WebSockets, Redis handles caching, and PostgreSQL powers persistence. Market data flows from yfinance and custom NSE scrapers.

This entire project came together through vibe coding—no heavy specs, just prompting ideas, iterating fast, debugging in real-time, and letting the architecture emerge organically through conversation. Pure creative flow
turned into production code.

Santa.com

Santa.com is a modern e-commerce platform for Indian retail with AI-powered
Features: Visual Search (snap-to-shop), AI Stylist, AR Try-On, and Squad Cart for social shopping. Includes Haggle mode for price negotiation, Flash Sales, coupon system, real-time order tracking, and a full admin dashboard with analytics, vendor KYC, and multi-role access.

Stack:
Mobile: (App and Backend)
Flutter (Riverpod, GoRouter, Supabase, Firebase, flutter_animate, Lottie), Node.js + Express, JWT auth, Supabase, PostgreSQL.

Mobile: Admin
Flutter (Riverpod, GoRouter, Supabase, Firebase, flutter_animate, Lottie), React + Vite, TanStack , Query, Tailwind, Framer .

Highlights:
- 200+ products, 10+ categories, smart search & filters
- Visual Search, AI Stylist, Squad Cart, Haggle
- Wishlist, Cart, Checkout, Order Tracking
- Admin: Analytics, Product/Order/Vendor management, Coupons, Audit logs, 2FA

This entire ecosystem—mobile app, backend API, admin panel—came together through vibe coding. No heavy specs, just prompting ideas, iterating fast, and letting architecture emerge organically. Creative flow turned into production code.

Tele Medicine App(HIPPA Compliant)

A HIPAA-compliant telehealth platform with three role-based mobile portals — Patient, Doctor, and Admin. Patients can sign up/log in with MFA, book video consultations by selecting a doctor and time slot, pay via Stripe, view medical records/vitals/prescriptions, track medications with reminders, and use an AI-assisted symptom checker. Doctors manage their schedule and waiting room, consult with patients over video, write prescriptions, and track earnings. Admins manage users, doctor verification, and platform analytics — all backed by a dedicated REST API layer.

Stack: Flutter, Dart, riverpod, dio, GoRouter, flutter_secure_storage

***R*oles and Responsibilities:**
- Mobile & Backend API Developer (sole Flutter developer; one of 4 backend developers on the mobile API layer
- Built the entire Flutter app end-to-end across all three portals (Patient, Doctor, Admin) 50+ screens
- Designed and implemented the full /api/mobile/* REST API layer (60+ endpoints: auth, patient, doctor, admin, shared) consumed by the app
- Implemented the complete patient appointment-booking flow — doctor/slot selection through Stripe checkout confirmation.
- Implemented secure, role-based authentication with email-OTP MFA and idle-session auto-logout
- Fixed navigation and loading-state bugs across the app
- Configured app icons, branding, and build flavors (dev/prod) for iOS and Android

Alankar

Alankar is an AI-powered Indian classical music education app for Hindustani music learners. It delivers real-time pitch detection with sub-cent accuracy, AI vocal coaching via Gemini, authentic virtual instruments (harmonium, tanpura, tabla), and comprehensive progress analytics—helping musicians master sargam exercises and raags through data-backed practice.

Features include a zero-latency harmonium engine with bellow physics, 40+ traditional raags organized by time-of-day, 100+ sargam patterns, pitch visualization, streak tracking with GitHub-style heatmaps, and personalized AI feedback explaining why corrections are suggested.

Tools: Flutter (Provider, fl_chart, Material Design 3) on the frontend. Audio powered by flutter_soloud (C++ engine) for zero-latency playback and FFT based pitch detection. Supabase handles auth and cloud sync, SQLite enables offline-first persistence, and Google Generative AI drives the intelligent vocal coach.

This entire project came together through vibe coding—no heavy specs, just prompting ideas, iterating fast, debugging in real-time, and letting the architecture emerge organically through conversation. Pure creative flow turned into production code.